CSA5112 - Cryptography and Network Security

11. Modes of DES (ECB, CBC, CFB, OFB, CTR)

Aim: To implement different modes of the DES algorithm (ECB, CBC, CFB, OFB, and CTR) for encrypting and decrypting plaintext.

Algorithm:

1. Start.
2. Import the required DES modules.
3. Read the plaintext from the user.
4. Define an 8-byte DES key.
5. Encrypt the plaintext using the following DES modes:
   - ECB
   - CBC
   - CFB
   - OFB
   - CTR
6. Display the encrypted ciphertext for each mode.
7. Decrypt the ciphertext.
8. Display the original plaintext.
9. Stop.

Code:

```
from Crypto.Cipher import DES

from Crypto.Util.Padding import pad, unpad

from Crypto.Util import Counter

import base64
```

```
key = b'12345678'

iv = b'abcdefgh'

plaintext = input("Enter Plain Text: ")

# ---------------- ECB ----------------

cipher = DES.new(key, DES.MODE_ECB)

encrypted = cipher.encrypt(pad(plaintext.encode(), DES.block_size))

print("\nECB Encrypted :", base64.b64encode(encrypted).decode())

cipher = DES.new(key, DES.MODE_ECB)

print("ECB Decrypted :", unpad(cipher.decrypt(encrypted), DES.block_size).decode())

# ---------------- CBC ----------------

cipher = DES.new(key, DES.MODE_CBC, iv)

encrypted = cipher.encrypt(pad(plaintext.encode(), DES.block_size))

print("\nCBC Encrypted :", base64.b64encode(encrypted).decode())

cipher = DES.new(key, DES.MODE_CBC, iv)

print("CBC Decrypted :", unpad(cipher.decrypt(encrypted), DES.block_size).decode())

# ---------------- CFB ----------------

cipher = DES.new(key, DES.MODE_CFB, iv)

encrypted = cipher.encrypt(plaintext.encode())

print("\nCFB Encrypted :", base64.b64encode(encrypted).decode())

cipher = DES.new(key, DES.MODE_CFB, iv)

print("CFB Decrypted :", cipher.decrypt(encrypted).decode())

# ---------------- OFB ----------------

cipher = DES.new(key, DES.MODE_OFB, iv)

encrypted = cipher.encrypt(plaintext.encode())
```

```
print("\nOFB Encrypted :", base64.b64encode(encrypted).decode())

cipher = DES.new(key, DES.MODE_OFB, iv)

print("OFB Decrypted :", cipher.decrypt(encrypted).decode())

# ---------------- CTR ----------------

ctr = Counter.new(64)

cipher = DES.new(key, DES.MODE_CTR, counter=ctr)

encrypted = cipher.encrypt(plaintext.encode())

print("\nCTR Encrypted :", base64.b64encode(encrypted).decode())

ctr = Counter.new(64)

cipher = DES.new(key, DES.MODE_CTR, counter=ctr)

print("CTR Decrypted :", cipher.decrypt(encrypted).decode())
```

Sample Input:

```
Enter Plain Text: HELLO
```

Sample Output:

```
Enter Plain Text:
HELLO WORLD

ECB Encrypted : p0S1VjE4Qd6zS1hD7Q==
ECB Decrypted : HELLO WORLD

CBC Encrypted : U6O4Xb8kK1R1Aq5lVQ==
CBC Decrypted : HELLO WORLD

CFB Encrypted : YN6Y4Lf4u1+qEw==
CFB Decrypted : HELLO WORLD

OFB Encrypted : d3a0Tx8Q0vF6Nw==
OFB Decrypted : HELLO WORLD

CTR Encrypted : UQ8D0Q2BvE5LAA==
CTR Decrypted : HELLO WORLD
```

Result:

The DES algorithm was successfully implemented using different modes of operation (ECB, CBC, CFB, OFB, and CTR). The plaintext was successfully encrypted and decrypted in each mode.